

Evaluating an Offline Detector Against Adversarial Prompt Attacks

A Reproducible Framework for Prompt-Attack Generation and Defensive Evaluation

github.com/poojakira/llm-redteam-framework - public, default branch main.

EVIDENCE POSTURE

Verified, scoped, reproducible

Claims are constrained to the repository evidence snapshot and stated limitations.

175

Tests passed

1

Tests skipped

94.30%

Statement coverage

1.00

Random-split F1

SYSTEM VIEW

Architecture and Threat-Control Flow



Research Contribution

A reproducible offline prompt-attack evaluation framework that reports both the stronger in-distribution result and the weaker novel-phrasing result, making the detector's generalization gap explicit instead of hiding it behind a random split.

Verified Evidence

EV-01 Grouped-template F1: 0.9714	EV-02 Novel-phrasing OOD F1: 0.7188
EV-03 OOD precision: 0.5897	EV-04 OOD recall: 0.92
EV-05 OOD false-positive rate: 64% (16/25)	EV-06 175 passed, 1 skipped.
EV-07 94.30% statement coverage; CI gate is 90%.	EV-08 Current benchmark values remain pinned:
EV-09 random split F1 1.00	EV-10 grouped-template F1 0.9714

Method and Control Path

01	Generate adversarial and benign prompt corpora across defined attack categories.
02	Train/evaluate the offline TF-IDF + Logistic Regression detector.
03	Compare random, grouped-template, structural-fixture, and novel-phrasing OOD evaluations.
04	Emit JSON/SARIF evidence for CI.
05	Keep live-model behavior claims out of scope unless separately measured.

Limitations and Residual Risk

- Offline detector evaluation does not prove whether a live LLM would follow an injected instruction.
- English/template coverage is finite.
- Structural fixtures are not equivalent to true OOD natural-language paraphrases.
- No real-world jailbreak-defense rate is claimed.

Reproducibility

Expected CI evidence: 175 passed, 1 skipped, 94.30% coverage.

Evidence Provenance

Claim discipline Metrics come from the repository's verified metrics and research brief.
Scope discipline Benchmarks are reported only at their documented fixture, dataset, or test scope.
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